Beyond the PHQ-9 Total Score: Geometric Decomposition Suggests Suicidal Ideation Heterogeneity Within Moderate Depression

Brian Riggleman
Independent Researcher, Owatonna, Minnesota
March 2026

Abstract

Current clinical instruments measure symptom frequency on a single axis. The PHQ-9 produces a number between 0 and 27. The Columbia Suicide Severity Rating Scale produces binary yes-no assessments. Neither tracks direction of change. Neither distinguishes between clinical presentations that have the same score but may require different treatment. This paper proposes geometric displacement from an individually-calibrated baseline (sigma) as a candidate clinical metric for tracking therapeutic progress, using the Geometric Affective Survey (GAS, Riggleman 2026r) which maps patient state to a three-dimensional coordinate defined by valence, activation, and intensity, formalized in companion papers (Riggleman 2026j, 2026k, 2026l, 2026m). A three-axis decomposition of NHANES 2021-2023 PHQ-9 data (n=5,455) demonstrates the subtype problem: among 455 respondents scoring in the moderate depression band (PHQ-9 = 10-14), four geometric subtypes show suicidal ideation rates ranging from 7.8% to 27.5%, a 3.5x difference not captured by the PHQ-9 total score. While this cross-sectional finding in a general population sample cannot establish causality, it suggests that substantial risk heterogeneity exists within severity bands that the total score does not distinguish. 132 respondents scoring exactly PHQ-9 = 10 occupy geometric locations spanning the full range of valence and activation axes, separating into four clinically distinct presentations from the same nine items. This paper describes the theoretical basis for the metric, demonstrates the subtype problem on federal health data, illustrates trajectory tracking, and outlines a validation protocol.

1. Introduction

1.1 The Resolution Problem in Clinical Assessment

A patient scores 15 on the PHQ-9. That means moderate-to-severe depression. The clinician adjusts treatment accordingly. Two weeks later the patient scores 14. Progress? Stagnation? Noise? The instrument has difficulty distinguishing meaningful change from measurement error at this resolution.

The deeper problem may be dimensional rather than just resolution. Two patients with a PHQ-9 of 15 can present very differently. One is withdrawn, flat, disengaged: predominantly cognitive-affective symptoms (anhedonia, depressed mood, low self-esteem) with low somatic activation. The other is anxious, agitated, sleepless: predominantly somatic-arousal symptoms (insomnia, psychomotor agitation, fatigue) with high activation. These are different conditions that may require different interventions. The score is identical. The clinical reality may not be.

1.2 The Trajectory Problem

The PHQ-9 measures state. It does not measure trajectory. A patient at 12 could be improving from 18 or deteriorating from 8. The number does not carry directional information. The clinician must remember previous scores and compute the change mentally.

The Columbia Suicide Severity Rating Scale has a different limitation. It measures ideation and behavior as present-or-absent at the time of assessment. It does not track approach. A patient who may be moving toward suicidal ideation but has not yet crossed the threshold is not visible to the instrument until they cross it. Whether an instrument that tracks approach trajectory could provide useful early information is an empirical question this paper proposes to test.

1.3 The Proposal

This paper proposes that geometric displacement from an individually-calibrated baseline, measured using the three-axis Geometric Affective Survey (Riggleman 2026r), could provide a clinical metric that addresses both the dimensional and trajectory limitations of current instruments. The three axes are formalized independently: valence (Riggleman 2026j), activation (Riggleman 2026k), and intensity (Riggleman 2026l), with the unified geometry described in Riggleman (2026m). The subtype problem is demonstrated on NHANES data in Section 3. The trajectory concept is illustrated in Section 4. The validation protocol is specified in Section 6.

2. Geometric Displacement as Proposed Clinical Metric

2.1 Establishing Sigma

Sigma is the patient's emotional home. Their default state. Where they sit in affective space when nothing is pushing them away from baseline. Sigma is not a healthy state or a target state. It is a descriptive baseline. A patient with chronic depression may have a sigma that sits at moderately negative valence. That is their home. Treatment may

aim to shift sigma itself, but the first measurement priority is knowing where sigma is so that displacement can be computed relative to it.

Sigma would be established at intake using the baseline questions from the GAS (Riggleman 2026r). The patient describes their typical emotional state across the three axes. The expectation is that this baseline would be stable over weeks to months for most patients. For patients in acute crisis, sigma may need to be re-established after stabilization. The stability assumption requires empirical validation.

2.2 Computing Displacement

At each session the patient would complete the GAS target questions. The current state is a coordinate in 3D affective space. Displacement would be the Euclidean distance from sigma to current state, expressed in standard deviations of the patient's own baseline variability.

The direction of displacement would tell the clinician which axis is driving the change. A displacement of 2.0 SD along the negative valence axis would be a different clinical situation than 2.0 SD along the activation axis. The specific thresholds that correspond to clinical significance would need to be established empirically through the validation protocol.

2.3 Trajectory Analysis

Displacement vectors across sessions would form a trajectory in affective space. The trajectory would have direction, speed, and curvature. A patient whose trajectory shows consistent movement toward sigma could be interpreted as improving. A patient whose trajectory shows movement away from sigma on the activation axis while valence stays flat could be developing anxiety on top of existing depression. A patient whose trajectory accelerates away from sigma could be deteriorating.

3. The Subtype Problem: NHANES PHQ-9 Evidence

3.1 Dataset and Decomposition

The National Health and Nutrition Examination Survey (NHANES) 2021-2023 cycle includes PHQ-9 depression screening data for 5,455 respondents with complete responses to all nine items. This is a nationally representative sample of the US population collected by the Centers for Disease Control and Prevention.

The nine PHQ-9 items map to three geometric axes based on their clinical content. The valence axis (cognitive-affective depression) is computed from items measuring anhedonia, depressed mood, and low self-esteem. The activation axis (somatic-arousal) is computed from items measuring sleep disturbance, fatigue, and psychomotor disturbance. The intensity axis (severity-magnitude) is computed from items measuring appetite changes, concentration problems, and suicidal ideation. The PHQ-9 total score sums all nine items into a single number. The geometric decomposition preserves the dimensional structure that the sum discards.

3.2 The PHQ-9 Distribution

Figure 1 shows the NHANES PHQ-9 data from both perspectives. The left panel displays the traditional PHQ-9 histogram with severity bands. Of 5,455 respondents, 723 (13.3%) screen positive for at least moderate depression (score 10 or above). The right panel shows the moderate band (PHQ-9 = 10-14, n=455) decomposed into four geometric subtypes, plotted on valence and activation axes. All 455 respondents score moderate depression. They occupy four distinct regions of the geometric space.

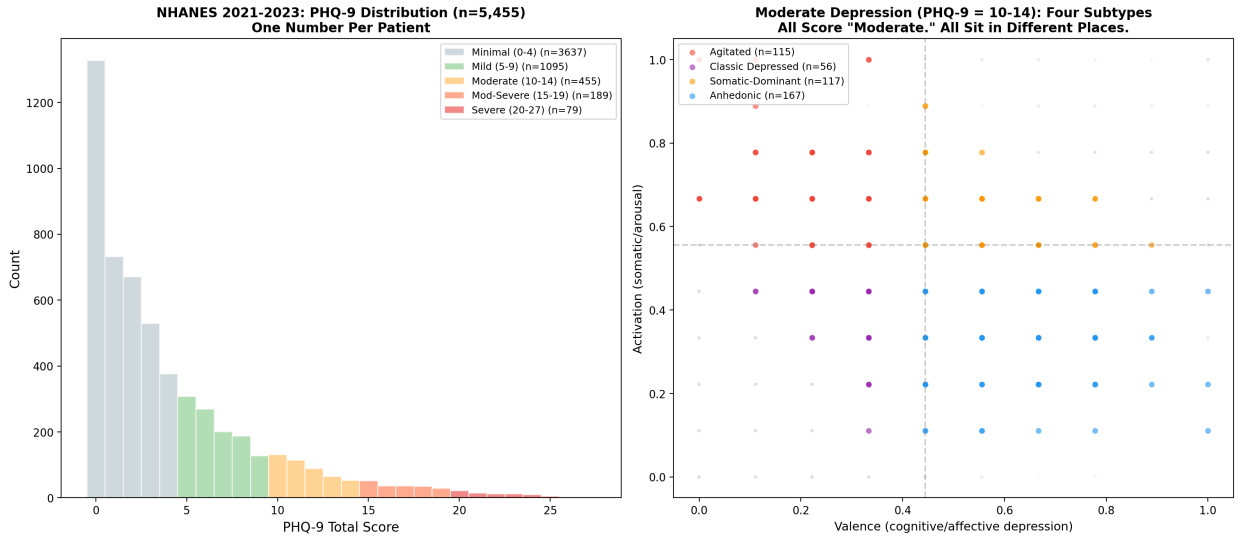

Figure 1. Left: PHQ-9 score distribution from NHANES 2021-2023 (n=5,455) with standard severity bands. Right: Moderate depression band (PHQ-9 = 10-14, n=455) decomposed into

four geometric subtypes. All patients score 'moderate.' They sit in four different geometric locations.

3.3 Four Subtypes Within Moderate Depression

The 455 respondents in the moderate depression band (PHQ-9 = 10-14) separate into four geometric subtypes with distinct clinical profiles:

Subtype	n	Valence	Activation	Intensity	Suicidal Ideation
Anhedonic	167	1.77	1.10	0.92	27.5%
Classic Depressed	56	0.92	1.18	1.38	19.6%
Somatic-Dominant	117	1.54	1.83	0.82	15.4%
Agitated	115	0.78	1.97	1.02	7.8%

The suicidal ideation rates range from 7.8% in the Agitated subtype to 27.5% in the Anhedonic subtype, a 3.5x difference within the same PHQ-9 severity band. While this cross-sectional demonstration in a general population sample cannot establish causality or trajectory, and suicidal ideation in NHANES is self-reported via a single item that may carry different clinical weight than in psychiatric settings, the finding suggests that substantial risk heterogeneity exists within the moderate depression band that the PHQ-9 total score does not distinguish.

The Anhedonic subtype, characterized by high valence-axis depression (loss of interest, depressed mood, low self-esteem) with low somatic activation, shows the highest suicidal ideation rate. The Agitated subtype, characterized by high somatic activation (sleep disruption, psychomotor disturbance) with lower cognitive-affective depression, shows the lowest.

This finding is consistent with clinical literature linking anhedonia and hopelessness to suicidal risk more strongly than somatic symptoms of depression. The contribution here is not the clinical finding itself but the demonstration that the three-axis decomposition can extract this clinically relevant distinction from the same nine items the PHQ-9 already collects, without adding any new questions.

3.4 Same Score, Different Geometry

Figure 2 shows 132 respondents who all scored exactly PHQ-9 = 10, the threshold for moderate depression. In the current clinical framework, all 132 receive the same clinical designation. In geometric space, they span the full range of both the valence and activation axes and separate into four distinct subtypes.

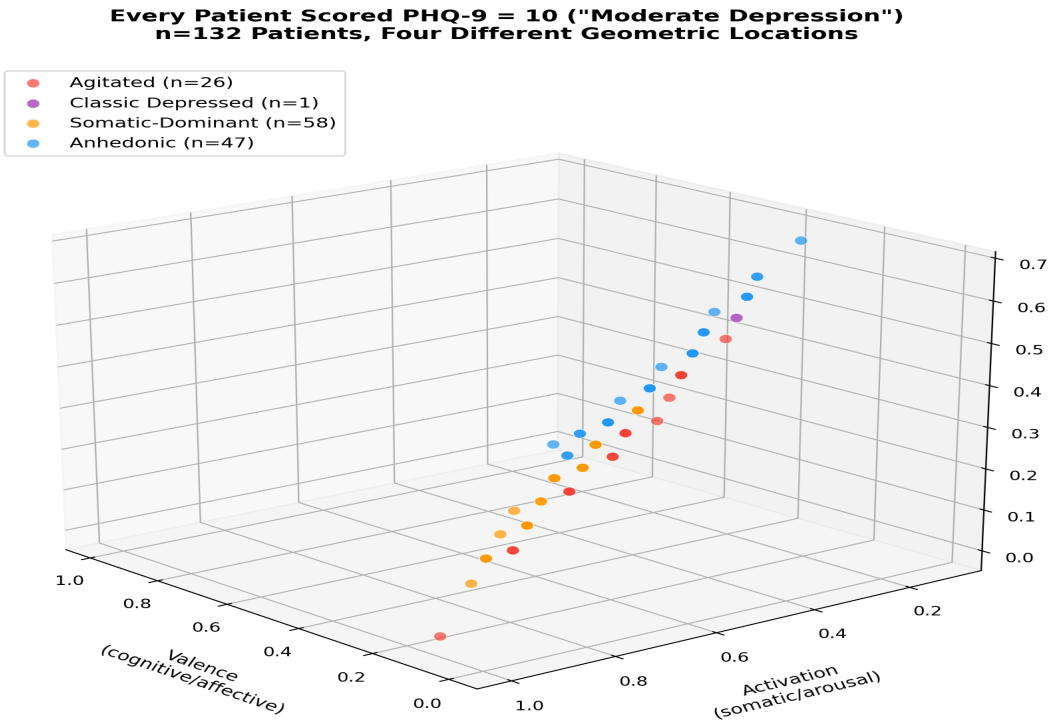

Figure 2. All 132 respondents scoring exactly PHQ-9 = 10 ('moderate depression') plotted in three-dimensional affective space. Same score. Four geometric locations. Four different clinical presentations from the same nine questions.

3.5 The Full 3D Distribution

Figure 3 shows all 5,455 respondents in three-dimensional affective space, colored by PHQ-9 severity category. The severity categories occupy different regions of the space but overlap substantially. The geometric view reveals the dimensional structure that the PHQ-9 total score collapses.

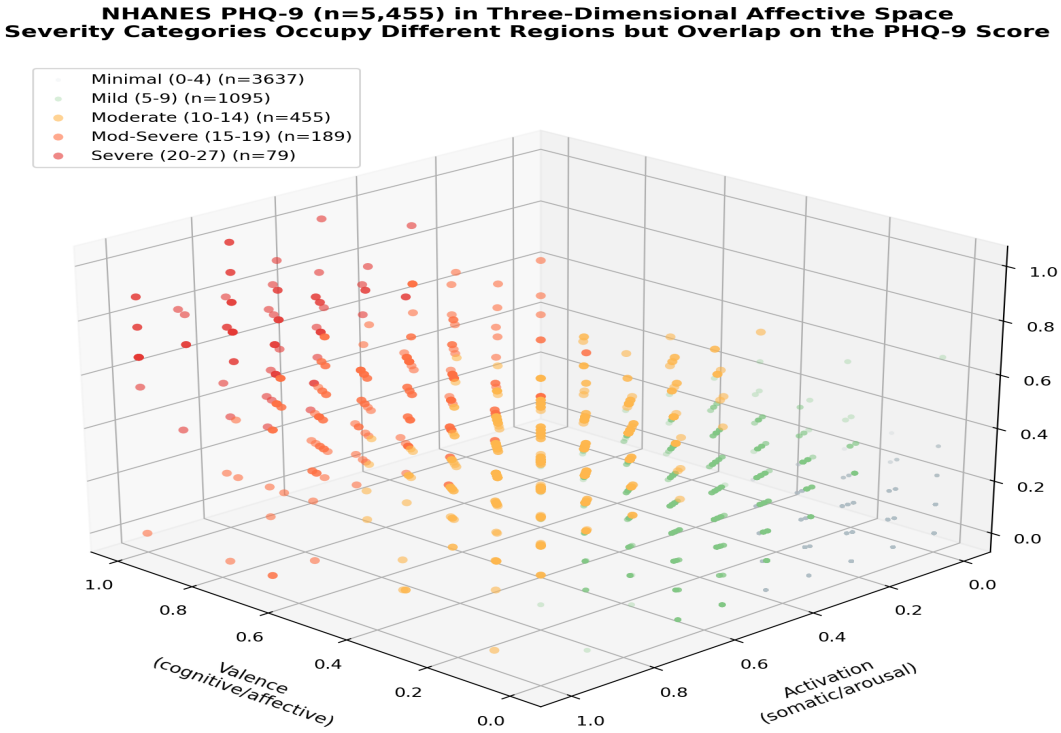

Figure 3. All 5,455 NHANES respondents plotted in three-dimensional affective space (valence, activation, intensity), colored by PHQ-9 severity category. The dimensional structure visible in the scatter plot is collapsed into a single number by the PHQ-9 total score.

4. Trajectory Tracking: Illustrated

4.1 Three Hypothetical Patients

To illustrate the trajectory concept, Figure 4 shows three simulated patients tracked over six sessions in three-dimensional affective space. Patient A (green) moves steadily toward sigma: improving. Patient B (orange) maintains approximately the same valence score but shifts dramatically on the activation axis: the single-axis view shows stability while the geometric view shows deterioration. Patient C (red) accelerates away from sigma on all three axes: rapidly deteriorating.

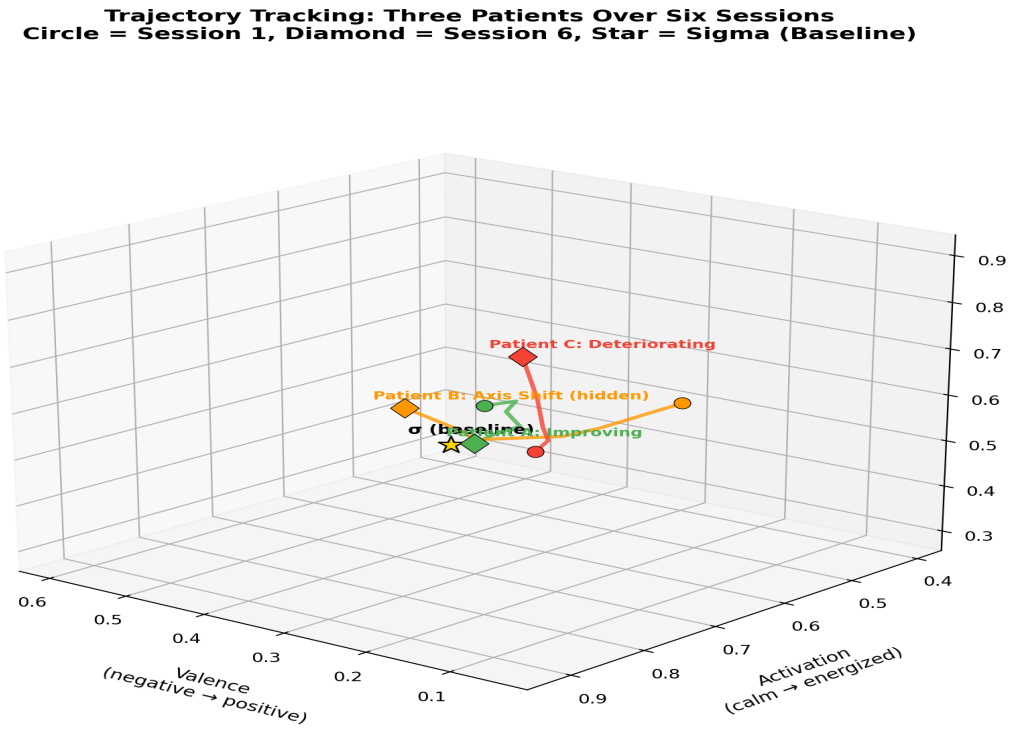

Figure 4. Simulated trajectories of three patients over six sessions. Gold star: sigma (baseline). Patient A improves. Patient B shifts axes without changing valence. Patient C accelerates away from sigma.

4.2 What the PHQ-9 Would See vs. What the Geometry Would See

Figure 5 presents the critical comparison. The left panel shows what the PHQ-9 would see: Patient A improving, Patient B stable, Patient C deteriorating. The right panel shows what geometric displacement would see: Patient B is deteriorating because activation is climbing while valence holds steady.

Patient B is the clinical case that matters most. The NHANES data in Section 3 shows why: in this sample, the Agitated subtype (high activation, lower valence depression)

and the Anhedonic subtype (low activation, higher valence depression) showed substantially different suicidal ideation rates. A patient shifting from Agitated toward Anhedonic presentation, maintaining the same total score, could be moving toward higher risk without the PHQ-9 total score registering any change. The geometric trajectory would capture this axis shift.

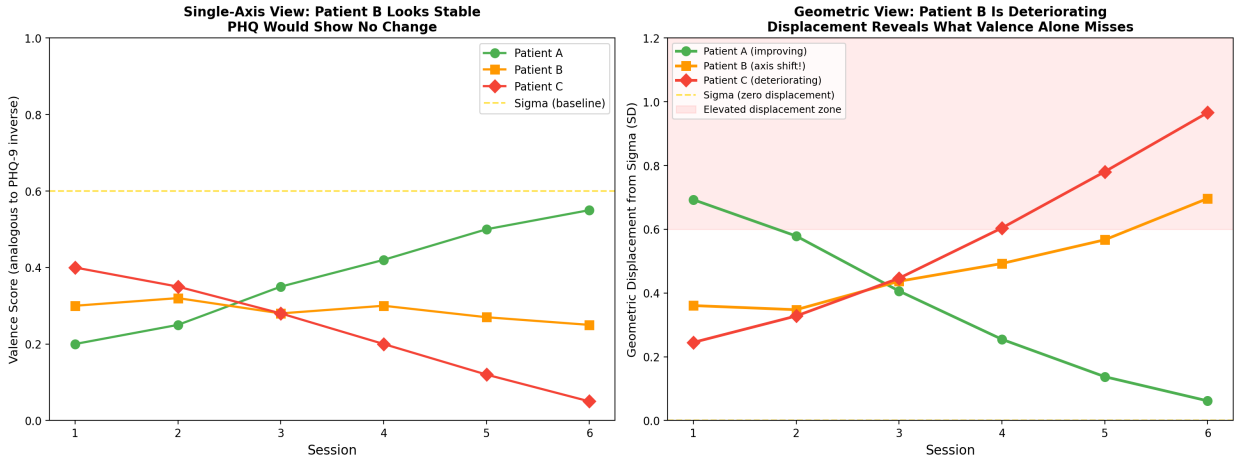

Figure 5. Left: Single-axis (valence only) view. Patient B appears stable. Right: Geometric displacement from sigma. Patient B is deteriorating on the activation axis. The geometry catches what the single axis misses.

5. Comparison with Existing Clinical Instruments

Feature	PHQ-9	Columbia (C-SSRS)	GAS + Displacement (proposed)
Dimensions measured	1 (total score)	2 (ideation, behavior)	3 (V, A, I) + sigma
Distinguishes subtypes	No	No	Yes (observed in NHANES)
Detects suicidal ideation risk differences within severity band	No (single score)	Binary only	Observed association (Section 3.3)
Tracks trajectory	Manual comparison	No	Proposed automatic
Individual baseline	Population norms	No baseline	Patient-specific sigma
Backward compatible	N/A	N/A	Yes (same 9 items, different math)

6. Proposed Validation Protocol

6.1 Concurrent Validity

Administer the GAS alongside the PHQ-9 and the C-SSRS at clinical intake for a cohort of patients presenting with depressive symptoms. Compute correlations between GAS valence displacement and PHQ-9 total score. The prediction is strong negative correlation with residual variance explained by the activation and intensity axes.

6.2 Discriminant Validity

Identify pairs of patients with identical PHQ-9 scores but different clinical presentations as assessed by structured clinical interview. Test whether the GAS places these patients at different coordinates. The NHANES demonstration provides proof of concept: respondents with identical PHQ-9 scores occupy different geometric locations with different suicidal ideation rates (Section 3.3). The prediction is that this pattern replicates in prospective clinical data with treatment outcome as the dependent variable.

6.3 Predictive Validity

Track GAS trajectories over a treatment course and test whether trajectory features (acceleration, direction change, multi-axis divergence) predict clinical outcomes better than PHQ-9 score changes alone. The specific prediction from the NHANES finding: patients whose geometric profile shifts from Agitated toward Anhedonic presentation (decreasing activation, increasing valence-axis depression) should show elevated suicidal ideation risk even if their PHQ-9 total score remains stable. This prediction is falsifiable and directly testable in longitudinal clinical data.

6.4 Sensitivity to Change

Compare the GAS displacement metric to the PHQ-9 change score for detecting treatment response. The prediction is that the GAS detects clinically meaningful change at earlier time points because it has higher dimensional resolution.

6.5 Clinical Significance Thresholds

The validation protocol should establish empirically calibrated thresholds by mapping GAS displacement values to clinician-assessed severity and to patient-reported distress levels. The NHANES data suggests that different geometric profiles within the same severity band may warrant different levels of clinical concern, given the observed variation in suicidal ideation rates.

7. Potential Early Warning Capability

The nightmare formation and trauma encoding paper (Riggleman 2026d) describes a trauma encoding threshold based on Al-Kaddah's Survival Tipping Point concept. The same threshold concept could potentially apply to clinical assessment.

A patient whose trajectory shows acceleration toward high displacement on multiple axes simultaneously could be approaching a critical state. The NHANES finding gives this proposal specificity: the Anhedonic subtype (high valence-axis depression, low activation, elevated intensity) showed the highest suicidal ideation rate in the data. A trajectory moving toward that subtype profile could potentially be detectable before the patient reports ideation. This remains hypothetical and entirely untested in longitudinal data. The NHANES cross-section can identify the destination that carries higher risk. Whether approach to that destination is detectable in real-time trajectory data is a separate empirical question.

This remains the most speculative application proposed in this paper. Whether trajectory-based approach detection is feasible depends on assumptions that require empirical validation. But the NHANES data establishes that the destination matters: not all moderate depression subtypes carry the same risk. If the geometry can detect movement toward the high-risk subtype, the early warning capability has a specific, data-supported target.

8. Giving Subjective Experience a Measurable Form

A person walks into a clinic and says they feel broken inside. The language of emotional suffering is subjective and can be minimized. A person who could point to a geometric displacement of 2.3 standard deviations from their baseline, in the subtype profile associated with 27.5% suicidal ideation in NHANES data, would have data to support their subjective report. Data is harder to dismiss than language.

The geometry would turn subjective experience into something that can be measured, compared, tracked, and communicated. The coordinates would be the coordinates. Whether this measurable form improves clinical communication and treatment decisions is a testable hypothesis. The NHANES evidence suggests that the additional resolution carries clinically relevant information.

9. Backward Compatibility

The NHANES demonstration establishes a critical practical point: the three-axis decomposition requires no new questions. The same nine PHQ-9 items, already collected in every primary care office, every mental health intake, and every federal health survey, produce the geometric coordinates when the math changes from

summation to decomposition. The information is already being collected. It is being discarded by the scoring method.

This means adoption does not require new instruments, new training, or new clinical workflows. It requires a different calculation applied to the same data. The existing PHQ-9 infrastructure, the forms, the electronic health record integrations, the training, the clinical guidelines, all remain unchanged. The scoring algorithm changes. The output changes from a single number to a three-dimensional coordinate. The nine questions stay the same.

A reasonable question is why this dimensional information has not been exploited in the 25 years since the PHQ-9 was developed. The answer is not that the field missed it. Factor analytic studies have consistently identified a two-factor structure in the PHQ-9 (Kroenke et al. 2010). The PHQ-9 scoring was designed to optimize reliability and clinical simplicity in primary care settings where a physician has minutes, not hours. A single number is fast to compute, easy to communicate, and straightforward to track. The tradeoff was dimensional resolution for practical usability. That tradeoff was reasonable when visualization tools were limited and clinical workflows were paper-based. The proposal here is that digital survey platforms and 3D visualization tools now make it practical to preserve the dimensional structure without sacrificing usability.

10. Ethical Considerations

Geometric displacement data is clinical data and subject to the same privacy protections as any clinical assessment. The 3D visualization must not be shared outside the clinical relationship without patient consent.

The suicidal ideation finding in Section 3.3 raises a specific ethical consideration. If the Anhedonic subtype shows substantially higher suicidal ideation than the Agitated subtype at the same PHQ-9 score, as the NHANES data suggests, clinicians using only the total score may not be capturing this risk differential. Whether this constitutes a meaningful gap in current clinical practice, or whether experienced clinicians already adjust for subtype through clinical judgment, is an empirical question the validation protocol should address.

The sigma baseline is a description, not a prescription. A patient with chronically negative sigma should not have that sigma treated as a target. The geometry describes where the patient lives, not where they should live. Treatment goals are set by the patient and clinician, not by the coordinate system.

11. Discussion

11.1 What This Paper Demonstrates

The PHQ-9 total score collapses the dimensional structure present in its nine items into a single number. The geometric decomposition preserves that structure. The nine items map to three clinical dimensions. Within the moderate depression band, four geometric subtypes show a 3.5x difference in suicidal ideation in NHANES data. This information exists in the data the PHQ-9 already collects. The total score does not capture it.

This paper proposes a metric and illustrates a trajectory concept. The displacement metric requires prospective clinical validation. The trajectory tracking requires longitudinal data. The NHANES demonstration is cross-sectional and cannot test trajectory predictions. What it can test, and does test, is whether the geometric decomposition reveals clinically meaningful variation within PHQ-9 severity bands. The answer is yes. The observed 3.5x difference in suicidal ideation within the moderate band warrants further investigation in clinical populations.

11.2 What Would Make This Useful

The metric would be useful if it provides clinically meaningful information that the PHQ-9 total score does not. The NHANES demonstration provides preliminary evidence that it does. The minimum threshold for clinical adoption is replication in a prospective clinical sample with treatment outcome data. The maximum value would be trajectory-based early detection of subtype shifts associated with elevated suicidal risk.

11.3 Limitations

The NHANES data is cross-sectional. It demonstrates the subtype problem at a single time point. It cannot demonstrate trajectory tracking or early warning capability. Those require longitudinal data from clinical populations.

The item-to-axis mapping is one reasonable interpretation based on clinical content. Other mappings could produce different results. Factor analytic studies of the PHQ-9 have consistently identified a two-factor structure (cognitive-affective and somatic) that aligns with the valence and activation axes used here, supporting the decomposition (Kroenke et al. 2010).

The suicidal ideation finding should be interpreted with appropriate caution. The NHANES sample is a general population survey, not a clinical sample. Suicidal ideation rates in clinical populations are higher, and the subtype distribution may differ. Replication in clinical samples is essential before any clinical application.

The instrument requires digital administration. Patients who cannot engage with a slider-based interface may not be able to complete the assessment.

12. Conclusion

The PHQ-9 gives a number. The geometry gives a location. Within the moderate depression band of the NHANES 2021-2023 PHQ-9 data, four geometric subtypes show suicidal ideation rates ranging from 7.8% to 27.5%. That 3.5x difference is clinically relevant, not captured by the total score, and extractable from the same nine items without adding a single question.

The instrument would not replace clinical judgment. It proposes to give clinical judgment better data. A coordinate instead of a number. A trajectory instead of a snapshot. Subtype identification that tracks to differential risk. Whether it delivers on these proposals in prospective clinical populations depends on the validation studies this paper is designed to motivate. The dimensional information is present in the data. The question is whether preserving it improves clinical outcomes.

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